

Software Engineering Lab 1

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Section: K

Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall route event budget proposals through a multi-stage approval workflow (Faculty Coordinator -> Finance Office -> Dean).	High	Pass: Status transitions sequentially upon digital sign-off. Fail: Request reaches Dean without Finance Office sign-off.	Ensures financial accountability and prevents unauthorized club spending.
FR-002	Functional	The system shall allow a Club Lead to submit an event proposal containing event details, expected attendance, and an itemized budget breakdown.	High	Pass: A submitted proposal is saved with status "Pending Faculty Review" and is visible in the Club Lead's dashboard. Fail: Proposal is accepted with a missing budget field.	Captures the minimum data needed to start the approval workflow.
FR-003	Functional	The system shall generate a unique, encrypted QR-code ticket for each attendee once an event's budget is fully approved and registration is confirmed.	High	Pass: Each issued ticket has a distinct QR payload tied to one attendee and one event ID. Fail: Two attendees receive an identical QR code.	Prevents ticket duplication and enables secure venue entry.
FR-004	Functional	The system shall allow Campus Admin to scan an attendee's QR ticket at check-in and mark it as "used", rejecting any ticket already scanned.	High	Pass: A second scan of the same QR code is rejected with a "Ticket already used" message. Fail: A used ticket is accepted on re-scan.	Stops duplicate entry and ticket sharing/fraud at venue check-in.
FR-005	Functional	The system shall allow a Club Lead to view a real-time budget dashboard showing requested, approved, and remaining funds for each event.	Medium	Pass: Dashboard values update within 5 seconds of a budget status change. Fail: Dashboard shows a stale total after an approval.	Gives club leads visibility to plan spending within approved limits.
NFR-001	Nonfunctional	The QR-code ticket scanner API shall validate entry authenticity in under 100 ms at venue check-in points.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Keeps entry lines moving during high-attendance events.

NFR-002	Nonfunctional	The system shall enforce role-based access control so that a Club Lead can view and manage only their own club's proposals, budgets, and tickets.	High	Pass: A Club Lead account attempting to open another club's proposal via direct URL/API receives a 403 Forbidden response. Fail: Cross-club data is returned.	Protects sensitive budget data and club privacy across the shared platform.
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Use-Case Flow Specification

Problem Statement #10 — Student Club Event Ticketing & Budget Portal

Use Case: UC-03 Request Budget Allocation (includes UC-04 Route Multi-Stage Approval)

Primary Actor: Club Lead

Supporting Actors: Faculty Coordinator, Finance Office, Dean (generalized as "Approver")

Preconditions

- Club Lead is authenticated and has an approved event proposal (UC-01) on file.
- The event proposal has not already been submitted for budget approval.

Postconditions

- Success: Budget request status is set to "Approved", and funds are reserved against the club's allocation.
- Failure: Budget request status is set to "Rejected", and the Club Lead is notified with a reason.

Main Success Scenario

1. Club Lead selects an approved event proposal and clicks "Request Budget Allocation".
2. System displays the itemized budget form pre-filled from the proposal.
3. Club Lead confirms the requested amount and submits the request.
4. System routes the request to the Faculty Coordinator and sets status to "Pending Faculty Review".
5. Faculty Coordinator reviews and digitally signs off on the request.
6. System forwards the request to the Finance Office and sets status to "Pending Finance Review".
7. Finance Office verifies fund availability and digitally signs off.
8. System forwards the request to the Dean and sets status to "Pending Dean Approval".
9. Dean gives final digital sign-off.
10. System sets status to "Approved", reserves the funds, and notifies the Club Lead.

11. Use case ends successfully.

Alternate Flow — 5a/7a/9a. Approver Rejects Request

5a1. The reviewing Approver (Faculty Coordinator, Finance Office, or Dean) rejects the request and enters a reason.

5a2. System sets status to "Rejected" and stops the approval workflow immediately (no further stage is notified).

5a3. System notifies the Club Lead with the rejection reason.

5a4. Club Lead may revise the budget request and resubmit, restarting the Main Success Scenario at step 1.